



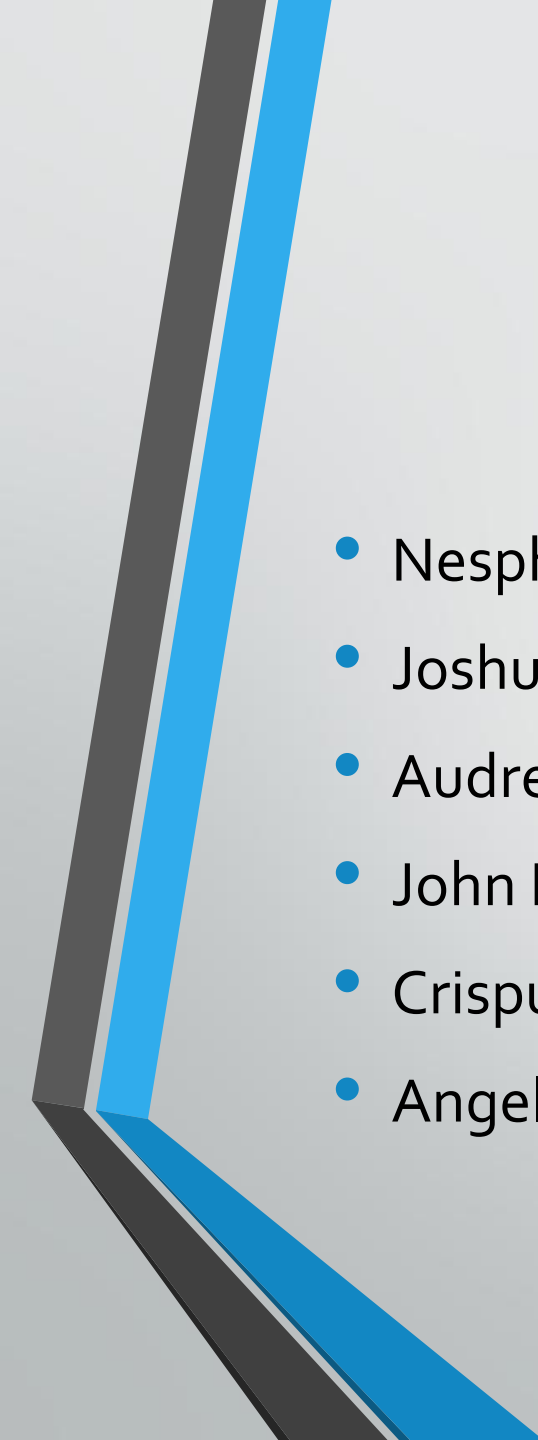
Couch-Potato Recommendation System

For:

Edge of the Seat Productions

By:

Data Munchers Consultants



Data Munchers Team

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Project Analysis Overview

We are tasked to build a recommendation system for a film production company. This analysis can be used to:

- Analyze user rating patterns to surface movies that align with their tastes
- Reduce the time spent browsing movie catalogues without finding suitable content.
- Increase user engagement by reducing decision fatigue.

Business Understanding

- Edge of the Seat Productions (ESP) wants to break into the streaming business and want a state-of-the-art recommendation system API that will keep the user engaged with quality recommendations based on previous movies watched and other similar users who have also seen the film.
- This way, they have a competitive edge over the current market leaders and offer value for money to their loyal users using tried and tested methods.

Data Understanding

- Our data was sourced from the [MovieLens](#) dataset from the GroupLens research lab at the University of Minnesota.
- This dataset contains around 100,000 ratings applied to 9,000 movies by around 600 users. This dataset was last updated in September 2018

grouplens

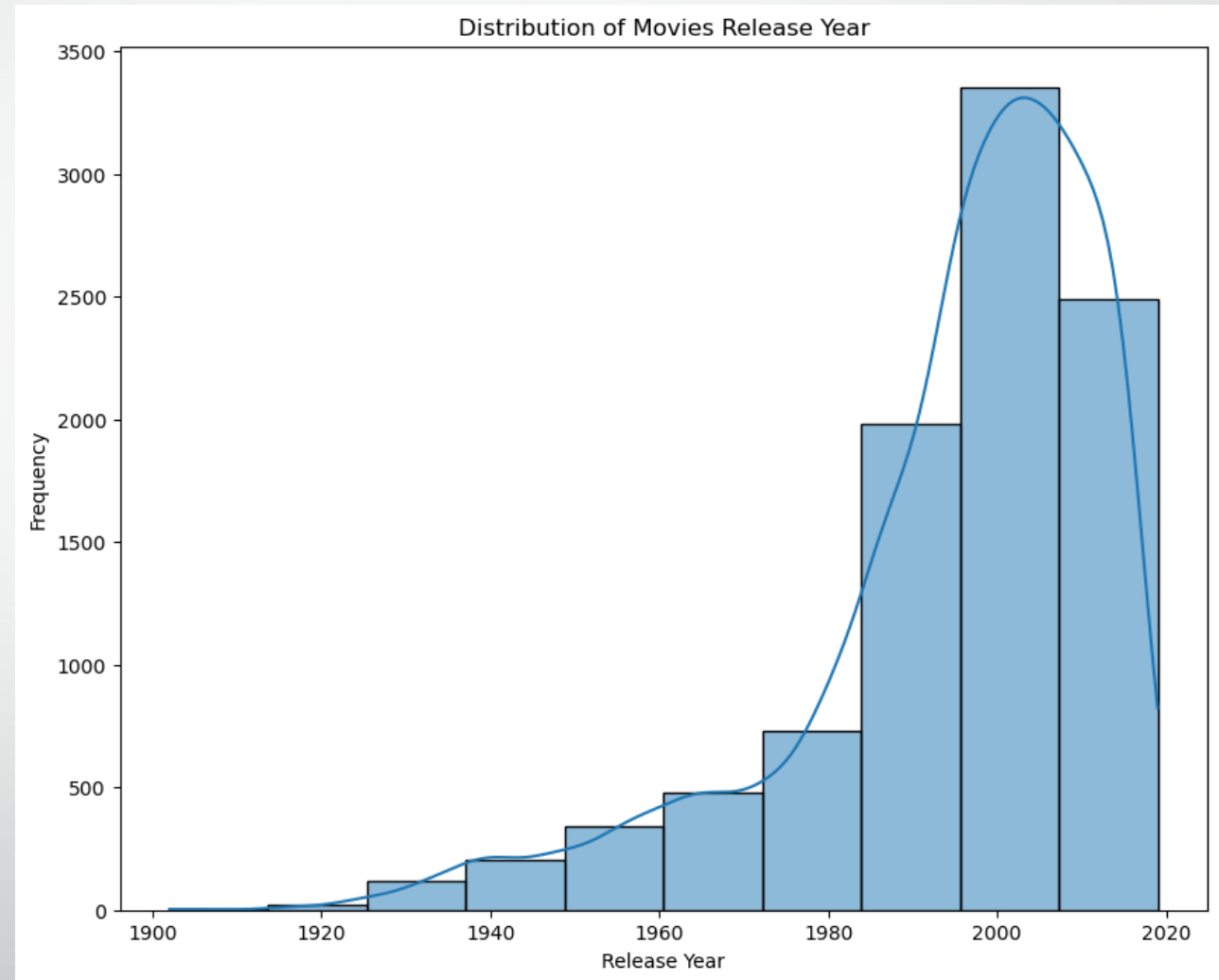
UNIVERSITY OF MINNESOTA



Data Analysis Report

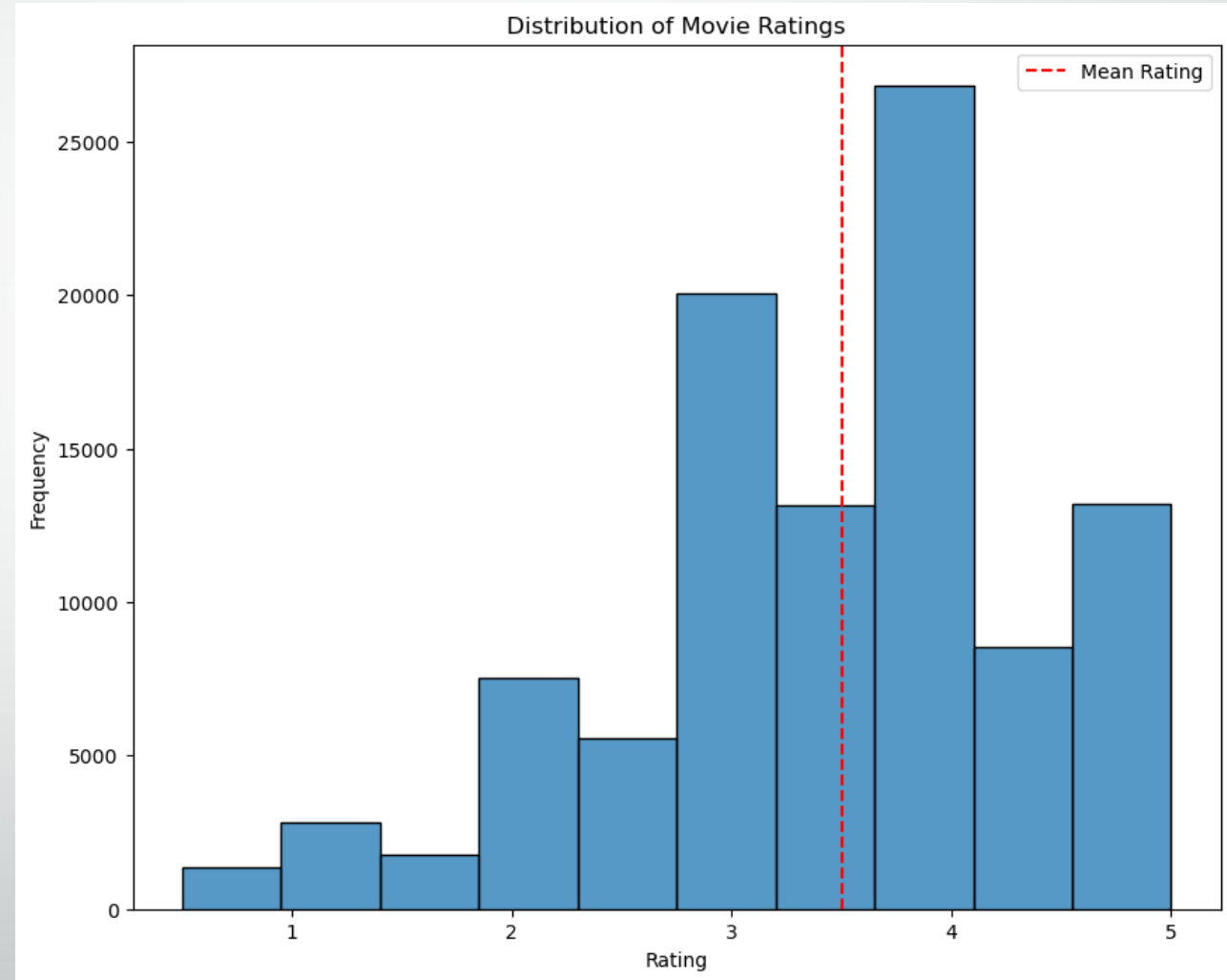
Data Analysis

- Most of the movies that were reviewed were released after early 1980s.
- The oldest movie in our dataset was released around the 1900s and the latest movie added into our dataset seems to be around 2018.



Data Analysis

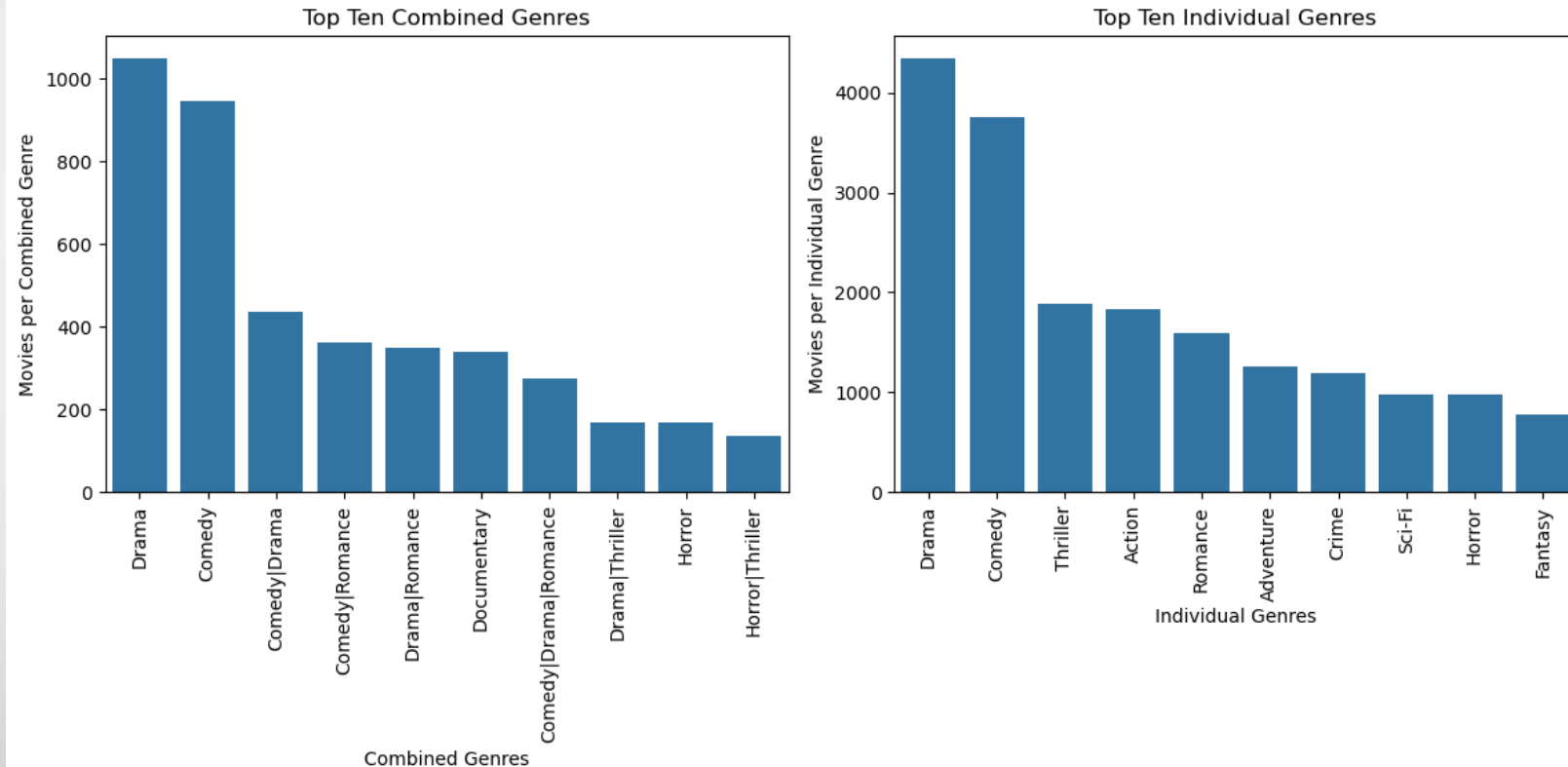
- Most of the movies were rated around 4 out of 5 stars.
- The mean rating in our dataset is around 3.5 out of a possible 5 stars.



Data Analysis

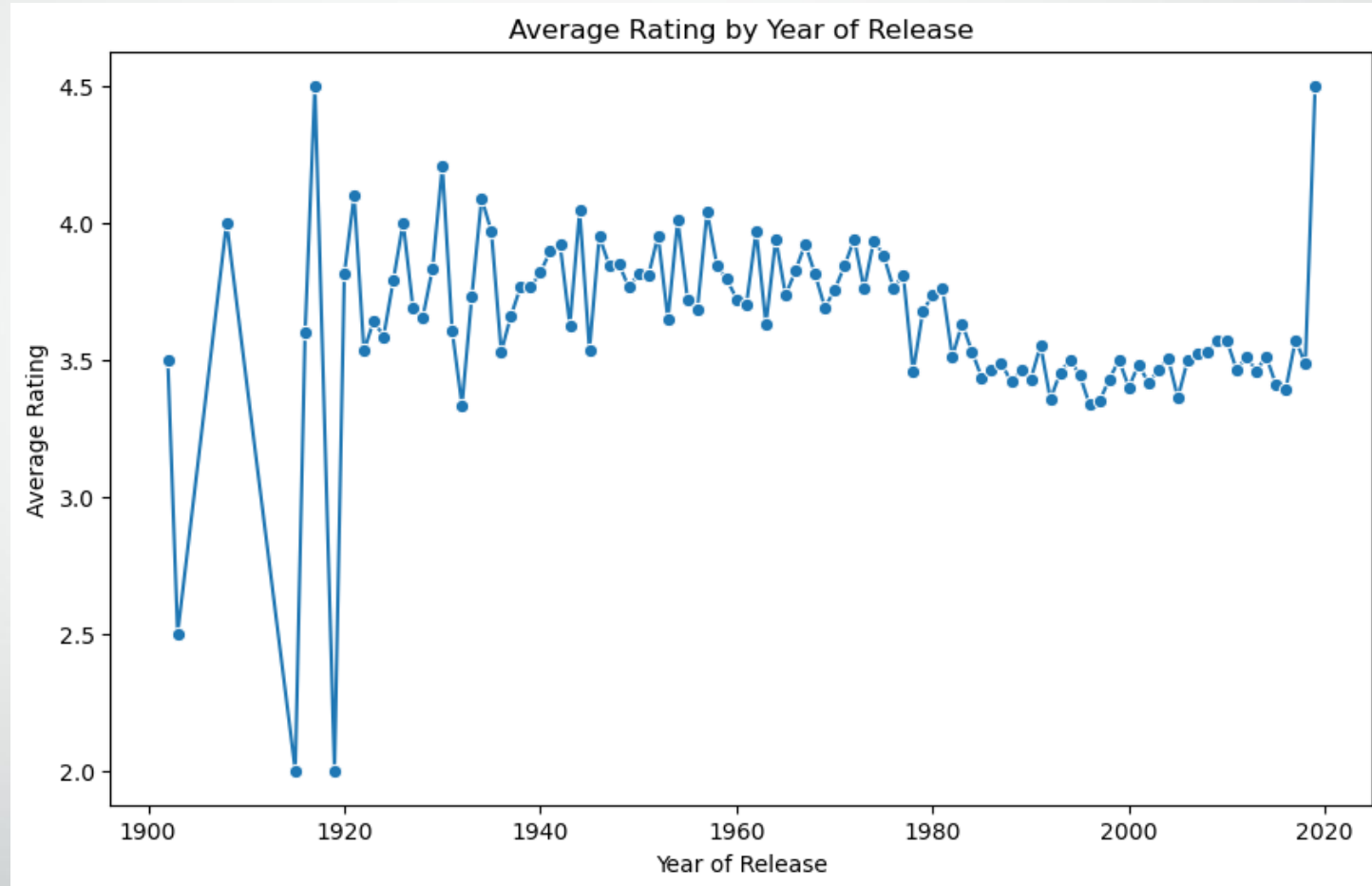
- Whether combined with other genres or individually, drama and comedy seem to be the leading genres.
- When combining, drama is still present in 4 other combinations followed closely by comedy which appears in 3 other combinations.
- A combination of Comedy and Drama is the third highest in the combined genres plot.

Genre Distribution



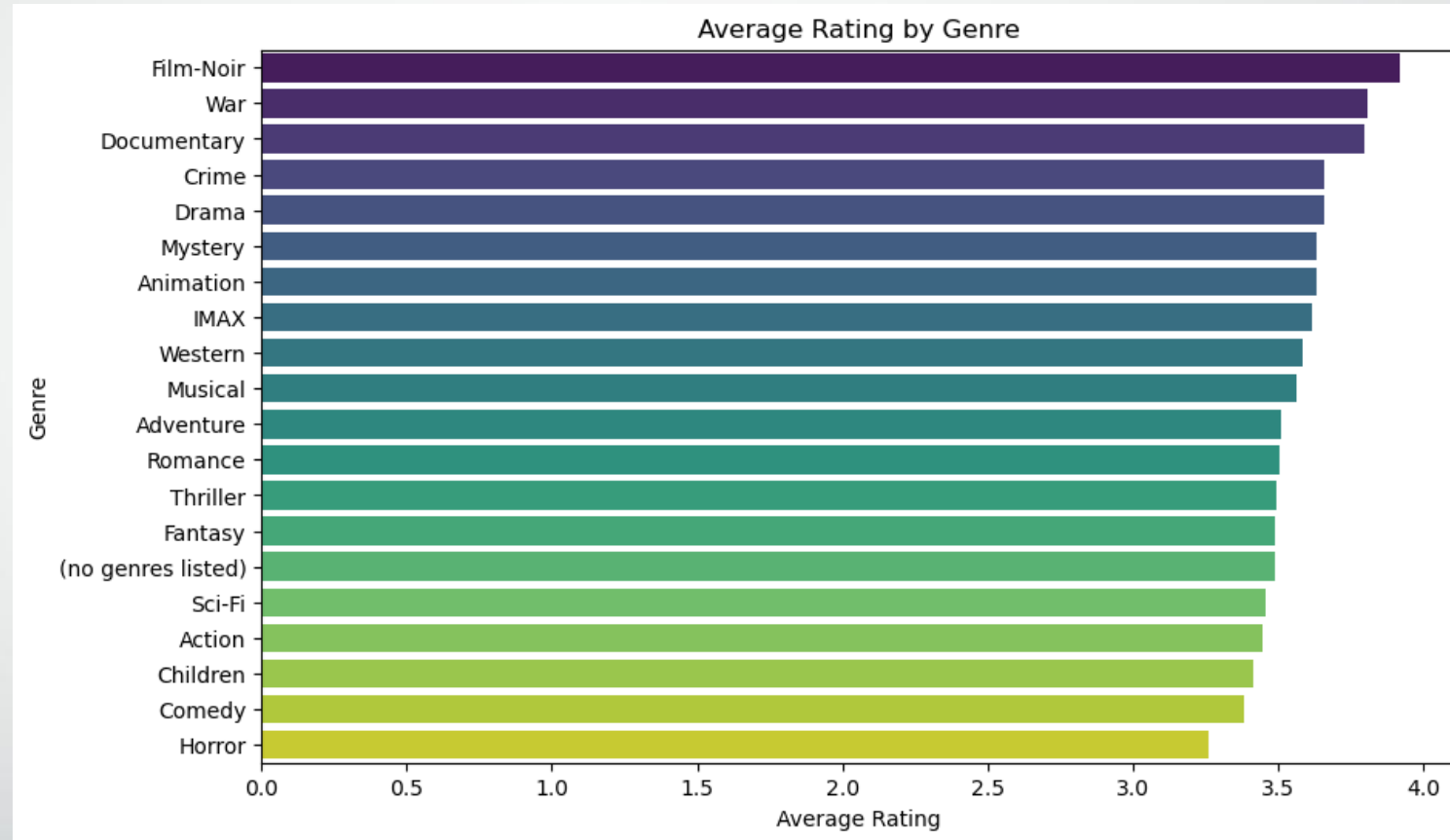
Data Analysis

- The lowest rating per year was realized around the late 1910s.
- Most movies released between 1940 and 1980 have an average rating of 3.75 stars.
- There seems to be a drop in rating after 1980 where most movies had an average rating of 3.5 stars. This may be as a result of lack of creativity and repetitive storylines.
- There was a sharp hike in 2020 to an average rating of 4.5. This may be due to technological advancements in the film making industry.



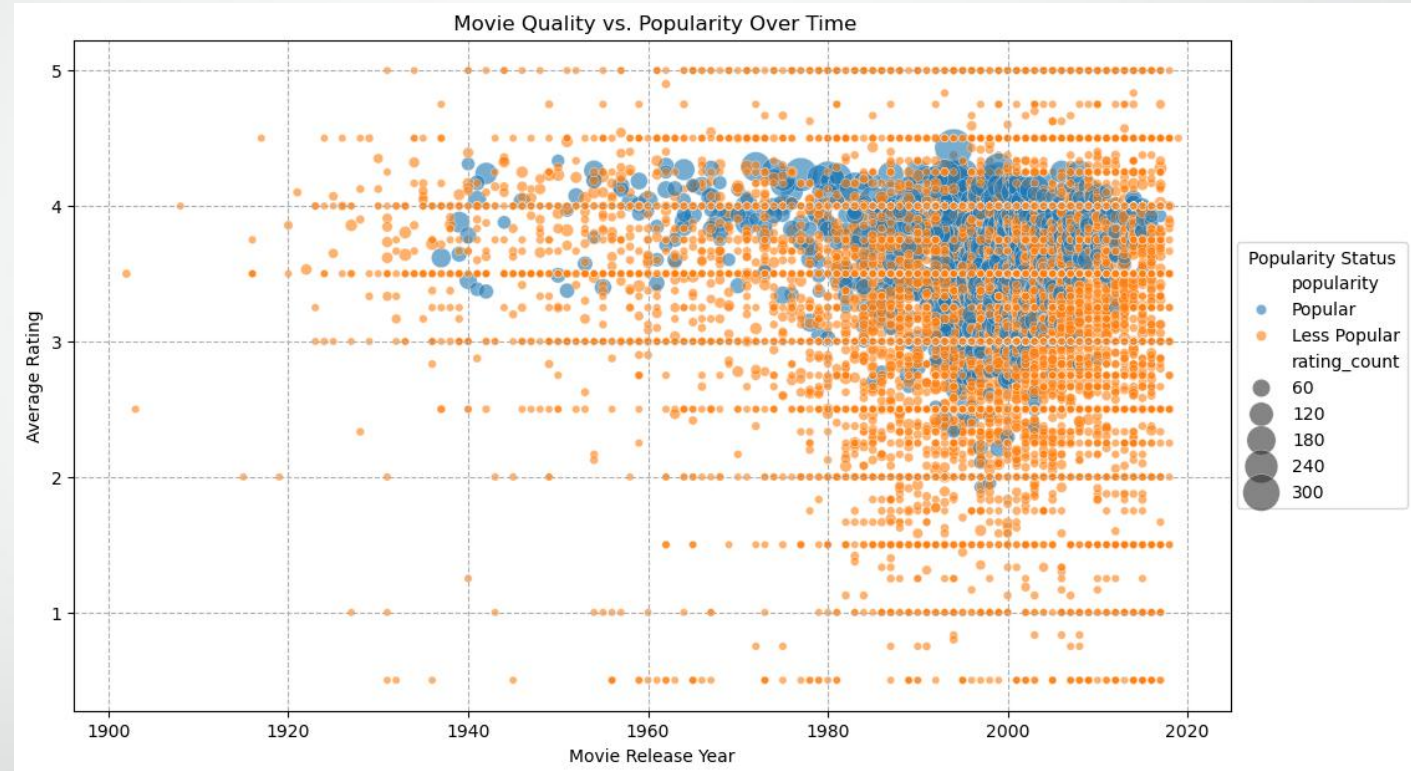
Data Analysis

- Film-Noir movies are black and white and majorly with no sound. They are rated the highest as they were very easy to follow and understand. It had an average of 3.9 stars.
- War, Documentary and Crime come 2nd, 3rd and 4th respectively to show that non-fictional films are rated highly.
- Horror was rated the lowest with around 3.2 stars
- There is a very small deviation between the top and bottom-rated genres



Data Analysis

- Recent decades show a much higher concentration of bubbles showing modern movies receive more ratings
- Very popular movies tend to cluster in later years and often have higher average ratings.
- Movies released before 1980 have a low rating count.
- A few older films appear as highly-rated & highly popular.



Minimum Viable Product (MVP)

- The module asks the user for a previous movie they liked. It then gives a recommendation of 5 other movies as well as the rating they would give.
- From our model, the predicted rating may deviate from the true rating by only 0.85 stars.

A screenshot of a web application titled "Couch-Potato Recommender System". The interface is dark-themed. At the top right, there is a "Deploy" button and a menu icon. The main heading is "Couch-Potato Recommender System". Below it, there is a prompt "Enter a movie you liked" followed by a text input field containing "liar liar". Below the input field is a button labeled "Get recommendations". Underneath the button, there is a section titled "Recommended Movies" with a small icon of two people. Below this title is a table with 4 columns: an index, "Recommended Movie", "Genre", and "Predicted Rating". The table contains 5 rows of data.

	Recommended Movie	Genre	Predicted Rating
0	George Carlin: You Are All Diseased	Comedy	4.66
1	Tampopo	Comedy	4.62
2	Louis C.K.: Live at the Beacon Theater	Comedy	4.62
3	Larry David: Curb Your Enthusiasm	Comedy	4.58
4	Everybody Wants Some	Comedy	4.57

Conclusion

- The oldest movie in our dataset was released around the 1900s and the latest movie added into our dataset seems to be around 2018. Most of the movies that were reviewed were released after early 1980s.
- The mean rating in our dataset is around 3.5 out of a possible 5 stars.
- Whether combined with other genres or individually, drama and comedy seem to be the leading genres.
- There seems to be a drop in rating after 1980 where most movies had an average rating of 3.5 stars. This may be as a result of lack of creativity and repetitive storylines.
- There was a sharp hike in 2020 to an average rating of 4.5. This may be due to technological advancements in the film making industry.
- Film-Noir movies are black and white and majorly with no sound. They are rated the highest as they were very easy to follow and understand. It had an average of 3.9 stars.
- War, Documentary and Crime come 2nd, 3rd and 4th respectively to show that non-fictional films are rated highly.

Recommendations

- ESP can bring back film-noir movies so that the new generation can also appreciate the steps taken in the film industry.
- ESP can also focus on non-fictional films as they seem to be among those rated highly from our dataset.
- ESP can put a wide variety of genres in there website as there is little or no deviation in ratings among individual genres.
- ESP should focus on movies released after 1982 to capture a majority of the population. However, they should still include even those before as some were still hits.
- ESP should also use a dataset that is always being updated so as to capture recent films and those that are still in production.

Future Steps

- Better algorithms and a neural network can be implemented to get finer results and reduce the deviation from 0.85 to less than 0.4 so as to get better recommendations and a better predicted rating. More experience, of course, is also needed to know where to improve and to increase domain knowledge in the sector.
- A larger and more modern dataset can be used to capture newer movies and even get a wider range of recommendations.
- Improving the user interface, where we even add the cover image of the film, a brief description of the film and how it has been rated by others like Rotten Tomatoes, TMDb and IMDb.



Questions?



Thank You!

Data Munchers Team

Click [here](#) to view the full project. Feel free to leave comments and insights.